INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	NC facility NC_way_186515922 -- station KMRN, America/New_York
Building	OSM 186515922 OSM way 186515922
Facade gap	not applicable -- one building, no second facade
Weather record	42,300 real hourly records from KMRN
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-03-19 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 11 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	7.305	18.0	7.110	0.987
01	mechanical	yes	switch budget	7.030	18.0	6.500	0.919
02	mechanical	yes	switch budget	6.585	18.0	5.560	0.874
03	mechanical	yes	switch budget	6.725	18.0	5.780	0.856
04	mechanical	yes	switch budget	4.805	18.0	2.610	0.798
05	mechanical	yes	switch budget	4.055	18.0	2.000	0.770
06	mechanical	yes	switch budget	3.615	18.0	0.780	0.750
07	mechanical	yes	switch budget	2.470	18.0	0.000	1.082
08	mechanical	yes	switch budget	6.580	18.0	5.220	1.683
09	mechanical	yes	switch budget	10.725	18.0	11.280	2.220
10	mechanical	yes	switch budget	13.305	18.0	15.220	2.580
11	mechanical	yes	switch budget	17.305	18.0	19.280	2.175
12	mechanical	no	dry-bulb	20.805	18.0	22.110	1.716
13	mechanical	no	dry-bulb	23.055	18.0	24.440	1.395

14	mechanical	no	dry-bulb	25.780	18.0	27.500	1.140
15	mechanical	no	dry-bulb	27.025	18.0	28.500	1.043
16	mechanical	no	dry-bulb	27.305	18.0	27.890	1.105
17	mechanical	no	dry-bulb	27.750	18.0	26.890	1.375
18	mechanical	no	dry-bulb	26.860	18.0	25.220	1.657
19	mechanical	no	dry-bulb	24.890	18.0	22.610	1.917
20	mechanical	no	dry-bulb	23.555	18.0	20.890	2.098
21	mechanical	no	dry-bulb	21.610	18.0	18.330	1.931
22	mechanical	no	dry-bulb	18.305	18.0	14.110	1.498
23	FREE	yes	-	16.890	18.0	12.780	1.081

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.305 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.030 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.585 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.725 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.805 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.055 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.615 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.470 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.580 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.725 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.305 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.305 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.805 C against a 18.0 C limit, so it fails by 2.805 C. It would take a limit 2.805 C higher, or a bound 2.805 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.055 C against a 18.0 C limit, so it fails by 5.055 C. It would take a limit 5.055 C higher, or a bound 5.055 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.780 C against a 18.0 C limit, so it fails by 7.780 C. It would take a limit 7.780 C higher, or a bound 7.780 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.025 C against a 18.0 C limit, so it fails by 9.025 C. It would take a limit 9.025 C higher, or a bound 9.025 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.305 C against a 18.0 C limit, so it fails by 9.305 C. It would take a limit 9.305 C higher, or a bound 9.305 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.750 C against a 18.0 C limit, so it fails by 9.750 C. It would take a limit 9.750 C higher, or a bound 9.750 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.860 C against a 18.0 C limit, so it fails by 8.860 C. It would take a limit 8.860 C higher, or a bound 8.860 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.890 C against a 18.0 C limit, so it fails by 6.890 C. It would take a limit 6.890 C higher, or a bound 6.890 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.555 C against a 18.0 C limit, so it fails by 5.555 C. It would take a limit 5.555 C higher, or a bound 5.555 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.610 C against a 18.0 C limit, so it fails by 3.610 C. It would take a limit 3.610 C higher, or a bound 3.610 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.305 C against a 18.0 C limit, so it fails by 0.305 C. It would take a limit 0.305 C higher, or a bound 0.305 C tighter, to change this hour.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.890 C, which is 1.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.081 C, and the margin is measured, not chosen: 1.081 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

WHAT IT IS WORTH OVER FIVE REAL YEARS, AT THIS SITE

Held-out days simulated	897 -- the agent never calibrates on a day it is scored on
Free cooling delivered	12.73 h/day, i.e. 4,649 h/yr
Chiller-hours avoided	+528.1 h/yr against a tuned reactive incumbent
12-hour plan stability	93.4 % of 21,084 re-plans change nothing at all

The ladder, one constraint at a time:

N-56-like: notice 0, skill 1.00, no constraint	+61.9 h/yr
+ switch budget 2, min dwell 3 h	+83.5 h/yr
+ dew-point gate 15 C (Green Grid WP#46 p.6)	+103.8 h/yr
+ notice 3 h, skill 0.50 (no perfect forecast)	+528.1 h/yr
+ unanchored, 4 measured FG offsets rotated	-113.2 h/yr

PRICED, IN THIS STATE'S OWN ELECTRICITY TARIFF

Chiller power	156.1 kW per MW of IT load (0.549 kW/ton, the ASHRAE 90.1-2019 minimum)
Electricity	8.72 cents/kWh (Virginia commercial, 2024 annual)
Energy avoided	82,443 kWh per MW of IT load per year
Value	$7,189 per MW of IT load per year

Everything above is PER MEGAWATT OF IT LOAD. This project has never measured a data centre's size and will not invent one; a reader who knows their own IT load multiplies once.

WHAT IS NOT CLAIMED

- The chiller COMPRESSOR only. Fans, chilled-water pumps, condenser pumps and cooling-tower fans keep running, and an airside economizer moves MORE air, so fan power can rise. The unmeasured term has the OPPOSITE SIGN, so this is an upper bound.
- Code-minimum chiller efficiency is the OPTIMISTIC end: Standard 90.1 is a floor, hyperscale plants beat it, and a better chiller saves less per hour switched off.
- Full-load kW/ton OVERSTATES the draw at the moment free cooling is available, because free cooling happens when it is cool and a chiller at a cool condenser runs below its rating. IPLV is the lower published number but is an annual-weighted metric under AHRI's assumed load profile, not the part-load point free-cooling weather sits at.
- The heat rejected is APPROXIMATED by the IT load. UPS, PDU and lighting losses inside the envelope add to it; some overhead sits outside the cooled space, so scaling by full PUE would overshoot. No PUE multiplier is applied.
- EIA STATE-SECTOR AVERAGES, not the site's tariff. A Loudoun County campus buys on a large-general-service contract that is not public.
- No carbon, no water, no maintenance and no demand charges. Tower water use moves the OTHER way when the chiller runs less.
- Every caveat on the hours is inherited: the headline is conditional on the bank sitting on the long facade (the refusal guard is priced at -3,124 h/yr where it fires) and the unanchored row is NEGATIVE. Those rows appear here with their signs intact.

HOW TO REPRODUCE EVERY NUMBER IN THIS REPORT

```
cd INTAKE-ARBITER/src && python run_all.py
```

That rebuilds every artefact from saved data and then audits it: 39 checks, 68 published figures re-read out of the files the code itself wrote, and it exits non-zero on any failure. It makes ZERO API calls -- every input is a saved FortyGuard response, a committed geometry file, or the station's own hourly record.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.